

Program -10

Develop a program to implement k-means clustering using Wisconsin Breast Cancer data set and visualize the clustering result.

Aim

The aim of this experiment is to implement the K-Means clustering algorithm on the Wisconsin Breast Cancer dataset in order to group the data points into clusters based on similarity and to visualize the clustering results using dimensionality reduction techniques such as Principal Component Analysis (PCA).

Theory

K-Means clustering is a widely used unsupervised machine learning algorithm that partitions a dataset into a predefined number of clusters (K) based on the similarity of data points. The main objective of the algorithm is to minimize the intra-cluster variation while maximizing the inter-cluster separation. It works by initially selecting K centroids, which may be chosen randomly, and then assigning each data point to the nearest centroid using a distance metric such as Euclidean distance. Once all data points are assigned, the centroids are recalculated as the mean of the points belonging to each cluster. This process of assignment and centroid update is repeated iteratively until convergence is achieved, that is, when there is no significant change in the centroid positions. The algorithm aims to optimize the following objective function:

$$J = \sum_{i=1}^K \sum_{x \in C_i} |x - \mu_i|^2$$

where (C_i) represents the set of data points in the (i^{th}) cluster and (μ_i) denotes the centroid of that cluster. Since the Wisconsin Breast Cancer dataset contains features with different scales, data standardization is performed to ensure that each feature contributes equally to the clustering process. Furthermore, as the dataset is high-dimensional, Principal Component Analysis (PCA) is applied to reduce the number of dimensions to two, enabling effective visualization of the clustered data. Thus, K-Means clustering helps in identifying inherent patterns and grouping similar data points without the use of labeled outputs.

Result

The K-Means algorithm was successfully applied to the dataset with **$K = 2$ clusters**.

- The data points were grouped into two clusters based on similarity in their features.
- After applying PCA, the high-dimensional data was reduced to two dimensions and visualized using a scatter plot.
- The visualization clearly shows separation between clusters, indicating effective grouping.
- The clusters approximately correspond to the actual categories (benign and malignant), although K-Means does not use label information.

```
[2]: # Import libraries
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Standardize data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply K-Means
kmeans = KMeans(n_clusters=2, random_state=42)
labels = kmeans.fit_predict(X_scaled)

# PCA for visualization
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Plot
plt.scatter(X_pca[:,0], X_pca[:,1], c=labels, cmap='viridis')
plt.title("K-Means Clustering")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.show()
```

